

Debugging Strategies

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Loops and Stopping

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Branching Logic

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Find a revealing case for Testing and debugging functions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Testing and debugging functions. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Test a new function independently

Define `stock(packs, loose)` to return `packs * 4 + loose`. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty `packs + loose` body.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Intended score is 3 per matching trait. Bug: score starts at 3 and adds 3 for each match. Give expected and actual results for 0,1,2 matches and repair the cause.

6. Solve the given case

Award two points per matching trait. A bug awards two for every trait.

Test one match and one nonmatch.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A loop should run four times but runs five. You change its colour and stopping condition at once. Which single change is relevant to test first?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 6 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

When a ranking surprises you, inspect feature values, weights, arithmetic, and sorting separately.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Intended score is 3 per matching trait. Bug: score starts at 3 and adds 3 for each match. Give expected and actual results for 0,1,2 matches and repair the cause.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Write expectations

Give expected results for remaining(6, 2), remaining(6, 6) and remaining(6, 0).

My reasoning and evidence

13. W2 Find a revealing case

A faulty function always returns 4. Which of your three cases pass and fail?

My reasoning and evidence

14. W3 Repair and retest

A faulty function adds capacity and used. For (6, 2), state actual and expected. Repair it and name a regression case.

My reasoning and evidence

15. W4 Test a new function independently

Define `stock(packs, loose)` to return `packs * 4 + loose`. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty packs + loose body.

My reasoning and evidence

16. W5 Limit the conclusion

All three stock tests pass. Does that prove correct handling of negative counts?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Debugging Strategies

A fictional shelf function should count books in complete bundles plus loose books.

Correct definition: shelf(bundle_count,loose) returns bundle_count*6+loose.

Faulty copied body returns (bundle_count+loose)*6. Predict tests before running.

AI ACADEMY / AI-2/6

Expected outputs versus a specific bug

	Bundles	Loose	Correct	Faulty
Test 1	2	1	13	18
Test 2	0	3	3	18
Test 3	2	0	12	12

Read a cell using both its row and column labels.

Debugging Strategies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Expected normal test

For (2,1), correct result is $2 \times 6 + 1 = 13$.



2. Faulty trace

Copied code gives $(2+1) \times 6 = 18$.



3. Zero-bundle test

For (0,3), correct 3; copied code 18.



4. Nonrevealing test

For (2,0), both return 12; a pass does not locate every bug.

CHANGE ONE THING

Repair the body to `bundle_count*6+loose`.

Explain the changed result using the matching lesson.

Which test most directly isolates the handling of loose books, and what should it produce?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Test the recommender's helper functions with normal, zero and boundary inputs before debugging the whole application.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Test the recommender's helper functions with normal, zero and boundary inputs before debugging the whole application.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional shelf function should count books in complete bundles plus loose books.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? For (2,0), both return 12; a pass does not locate every bug.

1. Zero-bundle test
2. Expected normal test
3. Faulty trace
4. Nonrevealing test

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Nonrevealing test For (2,0), both return 12; a pass does not locate every bug. Test cases need not all expose the same error. Here loose books must be added after multiplying bundles. A zero-loose case cannot distinguish the two formulas because multiplying zero loose books changes nothing. Keep expected and actual columns so an observed output does not quietly become its own expected answer.

Practice 2: explain the deciding evidence

Changed-case practice: Repair the body to $\text{bundle_count} * 6 + \text{loose}$. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Expected results for (2,1),(0,3),(2,0) are 13,3,12.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Expected results for (2,1),(0,3),(2,0) are 13,3,12. Only bundles are groups of six; loose books are counted individually.

Repair a missing building block

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?. Return to this week and explain what you changed.

Review Branching Logic (ai-2/3). Goal: Trace the selected branch in an if elif else chain. Short practice: Give expected labels for -1,0,10,11 and identify the equality boundary for Short.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
